Movie Rental Data Analysis – Project Guide

1. Project Objective

The objective of this project is to analyze the Movie Rental Database using SQL and Power BI to identify trends and insights related to:

- Movie rental revenue and payment trends
- Film performance and category popularity
- Customer behavior and rental patterns
- Customer segmentation and spending analysis
- Store performance and staff contribution
- Film inventory and availability

The project will involve SQL analysis for data exploration and Power BI for interactive visualization and reporting to evaluate business performance, understand customer preferences, and support data-driven decision-making.

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2. Dataset Overview

The dataset contains multiple related tables covering different aspects of the Movie Rental business, including films, customers, rental transactions, payments, staff, and inventory.
Main Tables

- **Film** – Movie details such as title, release year, rental rate, and rental duration.
- **Category** – Categories or genres of films, such as Action, Comedy, and Drama.
- **Film_Category** – Relationship between films and their categories.
- **Customer** – Customer information, including names, customer IDs, and store associations.
- **Rental** – Rental transaction details, including rental dates, return dates, and customer information.
- **Inventory** – Information about film copies available at each store.
- **Payment** – Payment transaction details, including payment amount, date, and related rental information.
- **Staff** – Staff information, including names and store associations.
- **Store** – Store details and relationships with staff, customers, and inventory.
- **Language** – Information about the languages available for films.

These tables are used to analyze rental revenue, film popularity, customer behavior, staff performance, store operations, and inventory management using SQL and Power BI.

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3. Data Model

The Power BI data model connects the Movie Rental tables using primary keys and foreign keys to enable accurate analysis across films, customers, rentals, payments, staff, and stores.

1. Film Analysis

Film → Film_Category → Category

The Film_Category table acts as a bridge between films and their categories.

2. Rental Analysis

Customer → Rental → Inventory → Film

This flow helps analyze customer rental behavior, film popularity, and inventory usage.

3. Payment Analysis

Customer → Payment

Rental → Payment

These relationships help analyze payment amounts, customer spending, and rental revenue.

4. Store & Staff Analysis

Store → Staff

Store → Inventory

Store → Customer

These relationships support store performance comparisons, staff contribution, customer distribution, and inventory availability.

5. Film & Actor Analysis

Film → Film_Actor → Actor

This flow helps analyze actors associated with films and explore film-related information.

6. Location Analysis

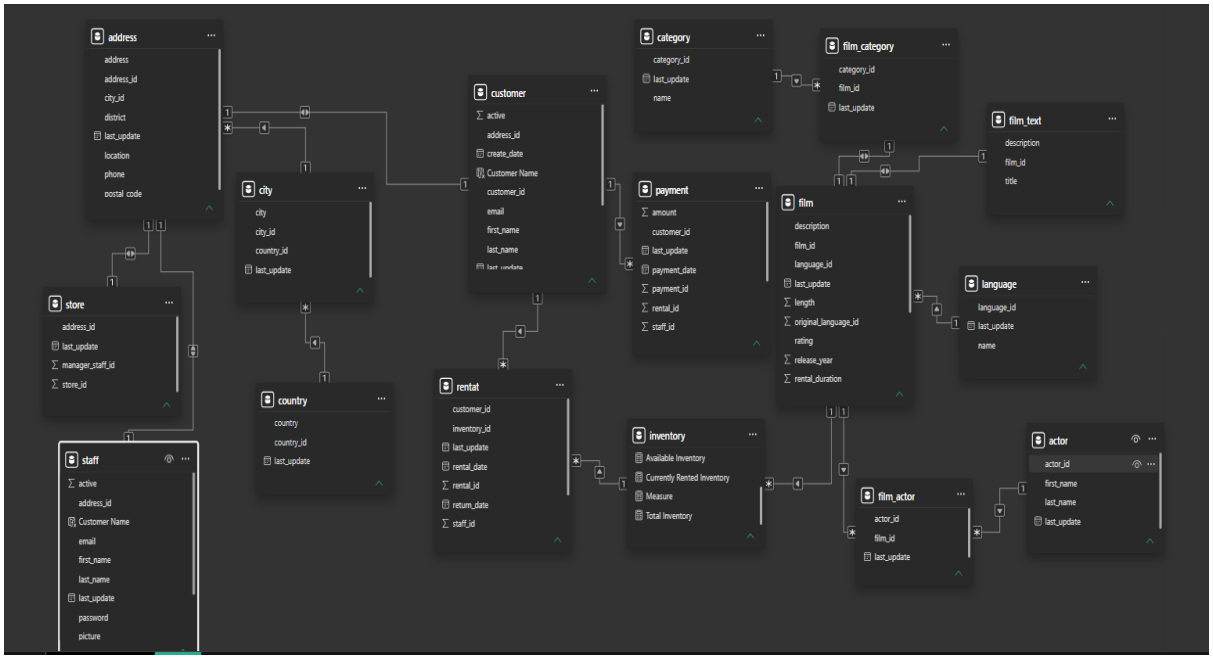
Address → City → Country

Store → Address

Customer → Address

Staff → Address

These relationships help analyze rental revenue, film category popularity, customer rental behavior, payment trends, staff performance, and store operations.



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4. SQL Analysis

1. **Customer Analysis**

- What are the purchasing patterns of new customers versus repeat customers?
- How does customer loyalty impact sales revenue over time?
- What are the demographics and preferences of the highest-spending customers?
- Do specific film categories attract different age groups of customers?

2. Film & Category Analysis

- Which films have the highest rental rates and are most in demand?
- Are certain language films more popular among specific customer segments?
- Are certain film categories more popular in specific locations?

- How does the availability of films in different languages impact customer satisfaction and rental frequency?

3. Store & Location Analysis

- Are there seasonal trends in customer behavior across different locations?
- How does the proximity of stores to customers impact rental frequency?
- What are the busiest hours or days for each store location, and how does it impact staffing requirements?
- What are the cultural or demographic factors that influence customer preferences in different locations?

- **Staff & Inventory Analysis**

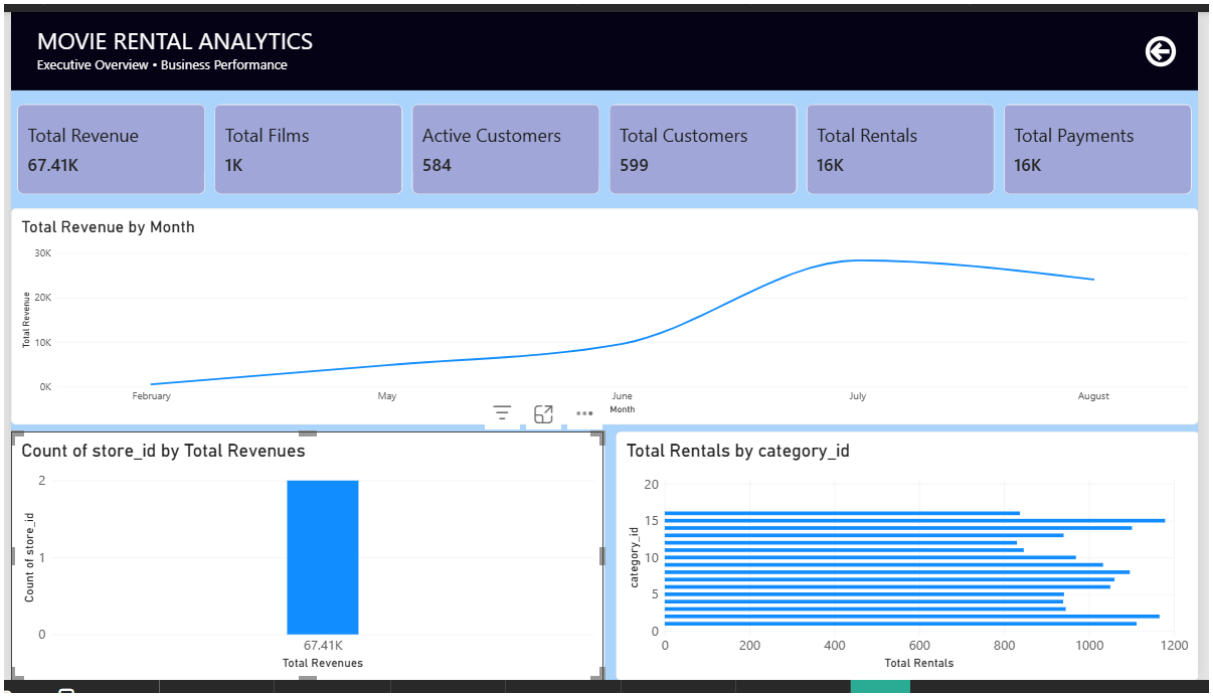
- **Are there correlations between staff performance and customer satisfaction?**
- **How does the availability and knowledge of staff affect customer ratings?**
- **How does the availability of inventory impact customer satisfaction and repeat business?**

5. Power BI Dashboard

The Power BI report is divided into the following pages:

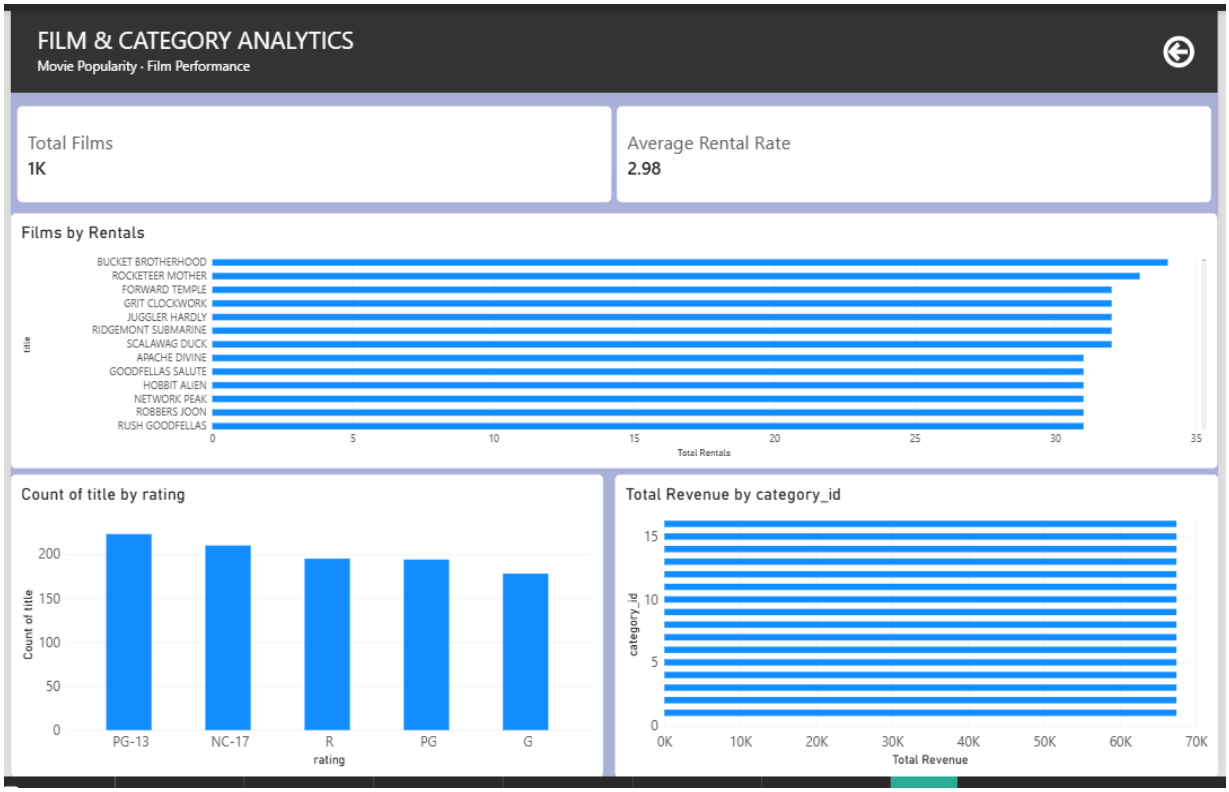
Page 1 – Movie Rental Overview

- Total Revenue – Total revenue generated from movie rentals.
- Total Rentals – Total number of rental transactions.
- Total Customers – Total number of unique customers.
- Total Films – Total number of unique films available.
- Revenue by Month – Monthly rental revenue trends.
- Revenue by Store – Comparison of revenue across stores.



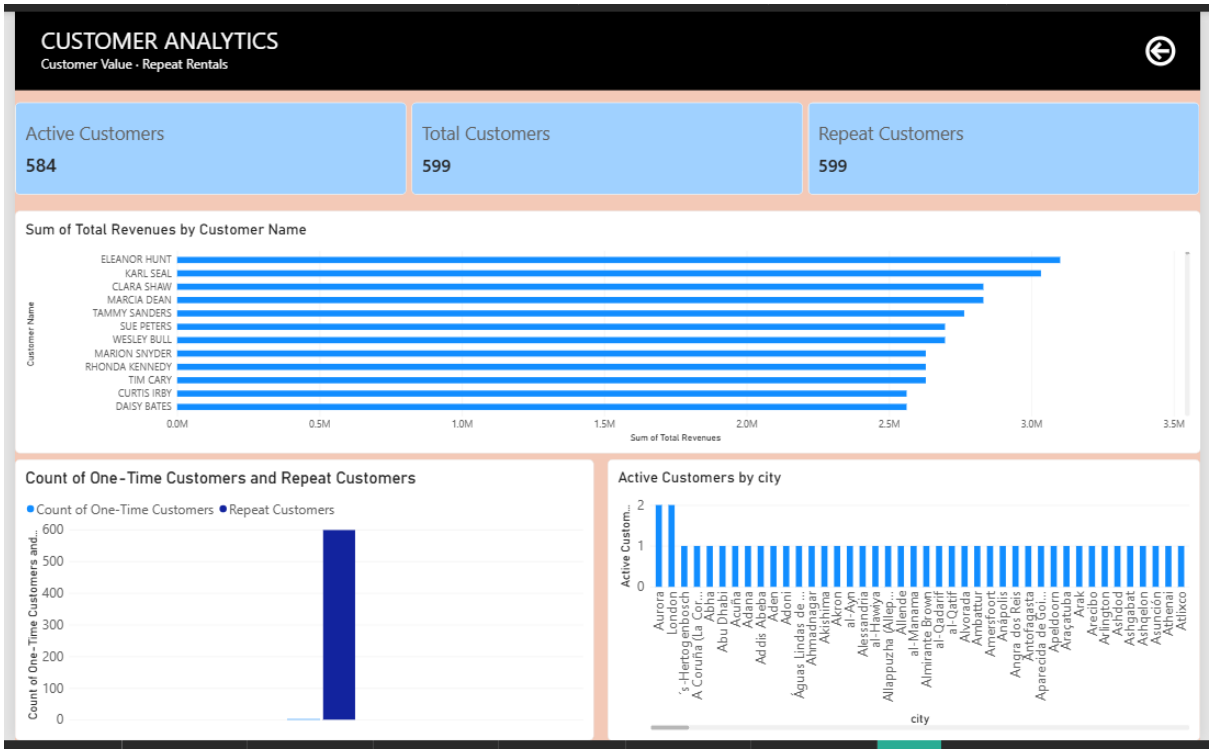
Page 2 – Film & Category Analysis

- Films by Category – Distribution of films across different categories, such as Action, Comedy, and Drama.
- Revenue by Category – Comparison of rental revenue generated by each film category.
- Rentals by Category – Number of rentals across different film categories.
- Rental Rates by Film – Comparison of rental rates across films.
- Film Rental Trends – Trends in film popularity and rental activity over time.
- Top-Rented Films – Identification of the most frequently rented films.



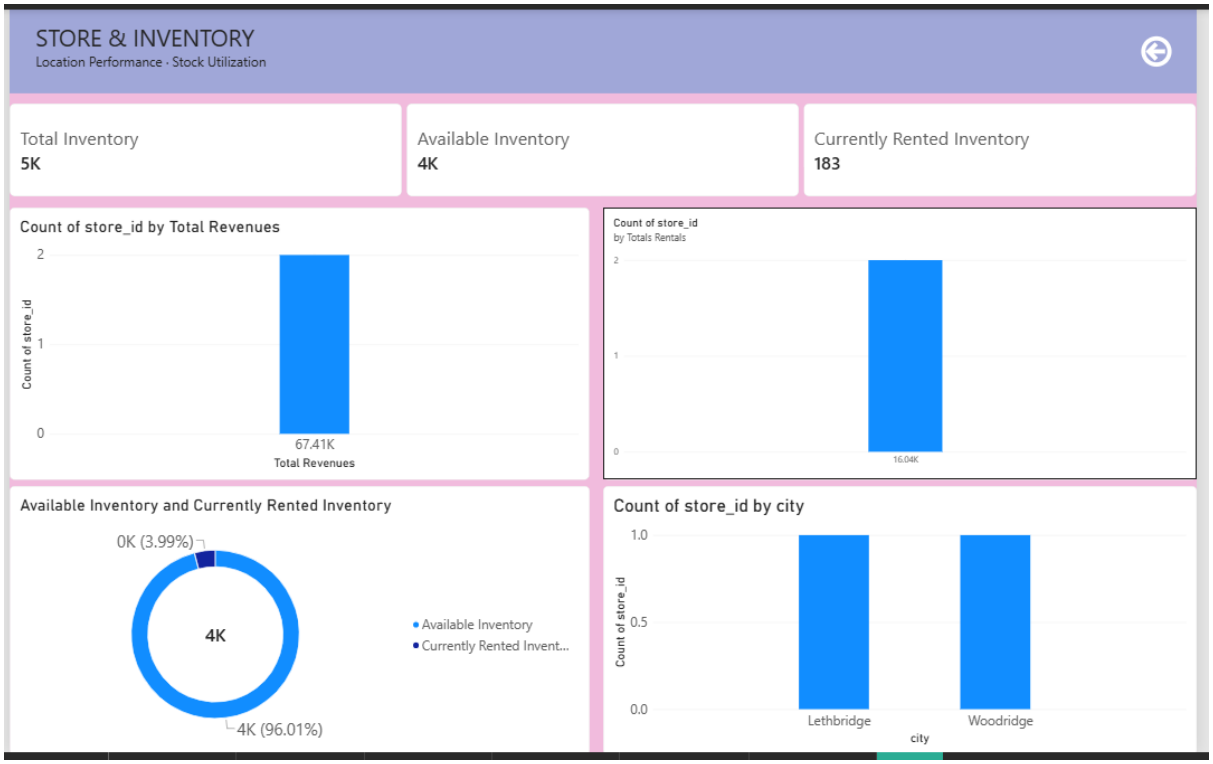
Page 3 – Customer Analysis

- Total Customers – Total number of unique customers in the database.
- New vs. Repeat Customers – Comparison of first-time and returning customers.
- Customers by Store – Distribution of customers across store locations.
- Customer Spending Distribution – Analysis of customer spending patterns.
- Customer Rental Trends – Changes in customer rental activity over time.



Page 4 – Revenue & Payment Analysis

- Total Revenue – Total revenue generated from movie rental transactions.
- Revenue by Payment Type – Distribution of revenue by payment method, if this data is available.
- Revenue by Film Category – Comparison of revenue generated by different film categories.
- Revenue by Store – Comparison of total revenue across store locations.
- Revenue Trends – Analysis of revenue changes over time.



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6. DAX & Interactivity — Movie Rental Project

Students will create appropriate DAX measures for KPIs and business analysis, such as:

- Total Rentals — Total number of rental transactions.
- Total Films — Total number of unique films.
- Total Customers — Total number of unique customers.
- Total Revenue — Total payment amount collected.
- Average Rental Rate — Average rental rate of films.
- Total Payments — Total number of payment transactions.
- Active Customers — Number of customers marked as active.
- Repeat Customers — Customers who have rented more than once.
- Revenue by Store — Total revenue generated by each store.
- Top Films by Rentals — Films ranked by rental frequency.
- Customer Revenue Contribution — Revenue generated by individual customers or customer segments.

- **Inventory Availability — Available inventory compared with total inventory.**

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7. Expected Final Outcome — Movie Rental Project

- The final Movie Rental Database Analysis and Power BI Dashboard Project should demonstrate the ability to:
    1. Understand and Work with a Relational Dataset — Understand movie rental data, including films, categories, customers, rentals, payments, inventory, stores, and staff.
    2. Write SQL Queries for Business Analysis — Use SQL joins, aggregations, subqueries, CTEs, and window functions to analyze rental trends, revenue, customer behavior, and film popularity.
    3. Build a Correct Power BI Data Model — Establish appropriate relationships between tables, handle duplicate and missing values, and ensure accurate filtering and aggregation.
    4. Create DAX Measures — Calculate KPIs such as total revenue, total rentals, total customers, total films, average payment, repeat customers, and store performance.
    5. Design Interactive Dashboards — Develop five dashboard pages:
        - Executive Overview
        - Film & Category Analysis
        - Customer Analysis
        - Store & Inventory Analysis
        - Payment & Revenue Analysis

- Identify Meaningful Trends and Patterns — Discover popular films and categories, monthly revenue trends, high-value customers, repeat rental behavior, and differences in store performance.
- Present Insights and Recommendations — Translate analytical findings into actionable recommendations, such as improving inventory allocation, promoting popular categories, retaining valuable customers, and identifying revenue growth opportunities.

- Validate Data and Ensure Accuracy — Cross-check SQL results against Power BI measures, verify relationships, and ensure that KPIs and visualizations display consistent results.
- Communicate Business Findings Professionally — Present the final dashboards with clear titles, appropriate charts, consistent formatting, interactive slicers, and concise business insights.

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8. Final Deliverables

Students should submit:

- **MECE Document**
- **SQL file** containing the analysis queries and use **Excel** to interpret results
- **Power BI (.pbix) file**
- **Project documentation/report**
- Key findings and recommendations

Bonus: Additional meaningful analysis beyond the given questions can be included.